Time-block forecast scores: AR(2) versus i.i.d. temporal pooling

Report on Experiments A and B (post `z.init` fix, assertion-guarded)

Score report

17 July 2026

Abstract

Two clean experiments compare the AR(2) skew-t model against the i.i.d.-in-time skew-t baseline under the time-block forecast protocol, by CRPS and by the Brier score at the q_{95} threshold, resolved by forecast lead time. **Experiment A** (setting 4, five expanding-window blocks × 5 datasets): the models tie once the reflected-ridge blocks are excluded; the only visible AR(2) gain sits at lead 1 and is gone by lead 3, exactly as the geometric memory of $\phi = (0.80, -0.35)$ predicts. **Experiment B** (block 1 only, guarded fits, datasets 1–10): at the same setting 4 the tie replicates; at the near-unit-root setting 5, $\phi = (0.15, 0.80)$, the AR(2) advantage becomes real and large — CRPS −31% over leads 1–5 (Wilcoxon $p = 0.040$; sensitivity $p = 0.010$–0.021) and a Brier improvement that is *flat in lead time* (AR(2) better at 14 of 15 leads, mean −0.008), consistent with a memory horizon ($h^* \approx 109$ days) that dwarfs the 15-day window. The advantage is a monotone function of temporal persistence, as the score-ceiling formula $\Delta\text{CRPS}/\text{CRPS} = 1 - \sqrt{1 - \text{SNR}\,\kappa_z(h)}$ requires.

1 Data-generating processes and design

All settings share one skeleton and differ *only* in the temporal dependence of the latent processes (`code/analysis/time_block_forecast/simstudy/setup.R`):

component	specification
marginal law	skew-t: $y_{st} = \mathbf{x}'\boldsymbol{\beta} + \lambda z_t + \varepsilon_{st}/\sqrt{\tau_t}$, $\tau \sim$ Gamma(6, 16) mixing (t-tails, $\nu \approx 6$), $\lambda = 3$ (right skew), $\beta_0 = 10$
skew latent	$z_t \mid \tau_t \sim \text{HN}(0, \tau_t^{-1/2})$, global ($K = 1$ knot, so the temporal signal lives entirely in τ, z)
spatial	Matérn $\nu = 0.5$ (exponential), range 1, spatial proportion $\gamma = 0.9$; 144 sites uniform on $[0, 10]^2$
temporal	Gaussian-copula **AR(2)** on $\tau^\star, z^\star$ ($w^\star$), coefficients ϕ per setting
record	$n_t = 200$ days; expanding-window blocks, each forecasting $H = 15$ days

id	label	$\phi = (\phi_1, \phi_2)$	spectral radius	memory h^* ($\varepsilon = .05$)
4	strong	$(0.80, -0.35)$	0.592	6 days (complex roots)
5	near-unit-root	$(0.15, 0.80)$	0.973	109 days (real roots, lag-2 dominant)

Protocol. Fit on a contiguous prefix $y[,1\!:\!T_o]$ (all sites observed), forecast the window $(T_o, T_o+15]$ with the seam-conditioned recursion of `forecast_block()`. Scores per lead h: CRPS (`crps_sample`, averaged over sites) and the Brier score for exceedance of the validation block's q_{95}, identical thresholds for both methods, paired by dataset. Both experiments post-date the `z.init` repair; all Experiment-B fits passed the pre-registered consistency guard (truth recovery $\wedge$ predictive spread; see the companion note `tex/guard_effect`), which replaced 13 of 40 attempt-0 chains. Experiment A excludes the four blocks flagged by the same assertions (reflected-ridge fits; `tex/lambda_phiz_ridge`).

2 Experiment A: setting 4, five blocks × five datasets

Expanding-window blocks with seams at $T_o = 50, 80, 110, 140, 170$; 21 clean (dataset, block) pairs.

Table 1: Experiment A, setting 4 (strong), clean 21 blocks. Mean score by lead; gap = AR(2) − i.i.d. (negative favours AR(2)).

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	1.822	1.514	−0.308	0.0277	0.0190	−0.0087
2	2.120	2.048	−0.072	0.0585	0.0552	−0.0034
3	2.273	2.288	+0.015	0.0509	0.0493	−0.0016
4	2.933	2.940	+0.008	0.0644	0.0644	+0.0000
5	3.330	3.318	−0.012	0.0977	0.0989	+0.0012
6	2.785	2.941	+0.156	0.0692	0.0801	+0.0109
7	2.285	2.261	−0.024	0.0518	0.0523	+0.0006
8	2.611	2.576	−0.035	0.0694	0.0701	+0.0007
9	1.968	1.991	+0.023	0.0298	0.0301	+0.0003
10	2.150	2.062	−0.088	0.0679	0.0603	−0.0076
11	2.055	2.085	+0.030	0.0403	0.0417	+0.0014
12	2.372	2.520	+0.148	0.0554	0.0608	+0.0053
13	3.087	3.232	+0.145	0.0712	0.0767	+0.0055
14	2.350	2.267	−0.083	0.0205	0.0231	+0.0026
15	2.432	2.298	−0.134	0.0768	0.0718	−0.0050
1–5	2.495	2.422	−0.074	0.0599	0.0574	−0.0025
1–15	2.438	2.423	−0.015	0.0568	0.0569	+0.0001

Reading. The AR(2) gain is visible exactly where geometric memory puts it: lead 1 (−0.308 CRPS, −0.0087 Brier, both the largest gaps in the table) and residually lead 2; from lead 3 onward the gaps oscillate around zero at the noise scale. Pooled over all leads the models tie (paired two-sided tests over the 21 blocks: Brier $p = 0.97$, CRPS $p = 0.78$). With spectral radius 0.592, the seam state is worth $\kappa_z(h) \approx 0.59^h$ of predictable signal: about a third at $h = 1$ and effectively nothing by $h = 5$ — the lead profile of Table 1 is that decay, read through two proper scores.

3 Experiment B: block 1, guarded, datasets 1–10

Block 1 trains on 50 days and forecasts days 51–65. Every kept fit passed the guard; setting-5 datasets 2 and 8 are excluded from the primary analysis (flagged; sensitivity below).

Table 2: Experiment B, setting 4 (strong), $n = 10$ datasets.

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	2.550	2.275	−0.275	0.0486	0.0552	+0.0066
2	3.169	3.141	−0.028	0.0923	0.0922	−0.0001
3	1.749	1.766	+0.017	0.0281	0.0236	−0.0045
4	1.861	1.841	−0.020	0.0414	0.0398	−0.0016
5	1.996	2.011	+0.015	0.0524	0.0540	+0.0015
6	2.244	2.259	+0.014	0.0333	0.0351	+0.0019
7	2.628	2.660	+0.032	0.0930	0.0938	+0.0008
8	1.775	1.827	+0.051	0.0358	0.0390	+0.0031
9	1.649	1.685	+0.035	0.0255	0.0262	+0.0007
10	2.172	2.143	−0.029	0.0936	0.0915	−0.0020
11	1.676	1.664	−0.012	0.0333	0.0347	+0.0013
12	2.132	2.096	−0.036	0.0774	0.0762	−0.0011
13	2.589	2.543	−0.046	0.1410	0.1368	−0.0043
14	1.805	1.808	+0.003	0.0311	0.0324	+0.0013
15	2.067	2.058	−0.009	0.0694	0.0682	−0.0013
1–5	2.265	2.207	−0.058	0.0526	0.0530	+0.0004
1–15	2.137	2.118	−0.019	0.0597	0.0599	+0.0002

4 Reading the tables

1. The advantage is a dose response in persistence. Pooled CRPS gap (leads 1–5): −0.074 (Exp. A, setting 4), −0.058 (Exp. B, setting 4), −0.844 (Exp. B, setting 5). Raising the spectral radius from 0.59 to 0.97 amplifies the advantage $\sim 14\times$, from undetectable to significant. This is what the ceiling formula $\Delta\mathrm{CRPS}/\mathrm{CRPS} = 1 - \sqrt{1 - \mathrm{SNR}\,\kappa_z(h)}$ demands: the seam state is only worth what $\kappa_z(h)$ retains of it.

2. The lead profile matches the memory, in both regimes. Setting 4 ($h^* = 6$ days): the gain lives at leads 1–2 and dies by lead 3, in *both* experiments (Tables 1 and 2, lead-1 CRPS gaps −0.308 and −0.275 — a clean replication across protocols). Setting 5 ($h^* = 109$ days): the gain does *not* decay inside the window — the Brier advantage is essentially flat, −0.006 to −0.010 at nearly every lead (Table 3), because a near-unit-root latent state is still informative 15 days out.

3. CRPS detects what Brier cannot yet. At setting 5 the two scores agree in direction (Brier better at 14/15 leads) but not in significance: CRPS is significant (W $p = 0.021$–0.040; sensitivity 0.010–0.021), Brier is not ($p \geq 0.12$). CRPS integrates the whole predictive distribution; the q_{95} Brier scores a single binary tail event per site-day and carries an order of magnitude more sampling noise per dataset. Powering the Brier endpoint to significance at the observed gap (≈ -0.008, between-dataset SD ≈ 0.02) needs roughly 30–50 paired datasets; extending the guarded run is mechanical (`run_block1_guarded.R --datasets=11:50`).

4. What the tie at setting 4 means. It is not evidence that AR(2) pooling is useless; it is the statement that at $h^* = 6$ days, 14 of the 15 scored leads lie beyond the memory of the process. Any study that pools such leads has diluted its own signal — which is why the setting-5 row, not

Table 3: Experiment B, setting 5 (near-unit-root), clean $n = 8$ datasets. AR(2) is better on CRPS at 12 of 15 leads and on Brier at **14 of 15** leads.

lead h	CRPS			Brier q_{95}		
	i.i.d.	AR(2)	gap	i.i.d.	AR(2)	gap
1	2.120	1.531	−0.588	0.0529	0.0437	−0.0092
2	3.366	1.879	−1.487	0.0651	0.0554	−0.0097
3	2.685	2.119	−0.567	0.0718	0.0719	+0.0001
4	2.788	1.552	−1.236	0.0258	0.0172	−0.0087
5	2.516	2.172	−0.344	0.0956	0.0876	−0.0080
6	2.717	1.979	−0.738	0.0285	0.0184	−0.0101
7	2.551	2.181	−0.371	0.0709	0.0563	−0.0146
8	2.845	2.121	−0.724	0.0470	0.0384	−0.0086
9	2.600	2.508	−0.092	0.0804	0.0706	−0.0098
10	3.786	2.736	−1.050	0.1091	0.1036	−0.0055
11	1.713	1.849	+0.135	0.0213	0.0120	−0.0093
12	3.225	2.402	−0.824	0.0806	0.0746	−0.0061
13	1.732	1.869	+0.137	0.0332	0.0260	−0.0072
14	2.739	2.347	−0.392	0.0512	0.0443	−0.0069
15	1.889	2.005	+0.116	0.0260	0.0180	−0.0081
1–5	2.695	1.851	−0.844	0.0623	0.0552	−0.0071
1–15	2.618	2.083	−0.535	0.0573	0.0492	−0.0081

more setting-4 data, was the correct next experiment.

Provenance. Experiment A: `simstudy/results/4-*.RData` (post-fix rerun), scored by `scores.R`; clean-block set from `fit_diag_4.RData`. Experiment B: `block1_positive_control/results/blk1-*.RData` (guarded driver), scored by the same protocol. Guard counterfactual: companion note `tex/guard_effect`.

Table 4: Paired one-sided tests (H_1: AR(2) better), per-dataset units. "Sens." keeps the flagged datasets ($n = 10$).

setting	endpoint	gap	t p	Wilcoxon p	sens. t/W p
4	CRPS, leads 1–5	−0.058	0.135	0.154	—
4	Brier q_{95}, 1–5	+0.0004	0.580	0.541	—
4	CRPS, 1–15	−0.019	0.216	0.270	—
4	Brier q_{95}, 1–15	+0.0002	0.546	0.500	—
5	CRPS, leads 1–5	−0.844	0.066	**0.040**	0.041 / **0.021**
5	Brier q_{95}, 1–5	−0.0071	0.232	0.417	0.224 / 0.305
5	CRPS, 1–15	−0.535	0.085	**0.021**	0.043 / **0.010**
5	Brier q_{95}, 1–15	−0.0081	0.119	0.221	0.124 / 0.380